Department of Computer Science & Engineering (CSE)
INDEPENDENT UNIVERSITY, BANGLADESH (IUB)
Mid Term Examination

Course Code: CSC 200	**Course Title:** Introduction to programming II	**Semester:** Autumn 2024
Full marks: 30	**Time:** 1 Hour 45 Minutes	**Date:** 07.11.204

- Write the answer in the blank space after each question
- No extra page/space will be entertained

Short Questions (2.5+2.5)

1. Justify the necessity of using loops in programming using an example.

Example:

2. Mention the differences between local and global variable using example.

Example:

Output Tracing (2.5+2.5)

3.
```
a = 99
b = '109'
c = 0
c = a + b
print ('Value of c is: ', c+=b)
```

Ans:

4.
```
def print_number(x):
   for i in range(x):
      print(i, end =' ')
   return

lines = 5

while lines > 0:
   print_number(lines)
   print()
   lines -= 1
```

Ans:

Error Tracing (3+3)

(Underline the line contains error and write correctly beside that line in question 4 and 5)

4.
```
import math
def primeFactors():
        while n % 2 == 0:
                print 2
                n = n / 2

        a = int(math.sqrt(n))
        for i in range (3, a+1, 2):
                while n % i = 0:
                        print i
                        n == n / i
        if n > 2:
                print n
primeFactors(n)
```

5.
```
def compute_gcd(x, y):
        while(y):
                x = y
                y = x % y
        return x

def compute_lcm(x):
        lcm = (x*y)//compute_gcd(x)
        return

num1, num2 = 54, 24
print("The L.C.M. is", compute_lcm(num1, num2))
```

Code Refactoring (5)

6. Replace while loop with the for loop and for loop with the while loop

```
def LCM(a, b):

    greater = max(a, b)
    smallest = min(a, b)

    for i in range(greater, a*b+1, greater):
        if i % smallest == 0:
            return i

    a, b = 12, 15
    print("LCM of", a, "and", b, "is", LCM(a, b))
```

Ans:

```
def is_prime(num):

    if num <= 1:
        return False
    i = 2
    while i * i <= num:
        if num % i == 0:
            return False
        i += 1
   return True
```

Ans:

Program Writing

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7. Write a function that takes input of N numbers as parameters representing marks obtained in quizzes. The function should calculate and return the average of best N-1 quizzes. Use variable length arguments to define the function.

8. Write a program that calculates discounted payable amount on purchase along with associated customer information. Take necessary inputs to run the program successfully. Define the separate function for the following list of tasks:
    - Retrieve the customer's name in the first function
    - Retrieve the customer's mobile number in the second function
    - Calculate the total purchase amount of the customer for N numbers of purchased elements in the third function
    - Calculate the payable amount after discount where the amount of the product with minimum price will be deducted from the total purchase amount in the fourth function
    - Finally print all information in the fifth function